

Final Project

Quantitative Economics, Fall 2025

February 2, 2026

Submit your solutions until March 6th 11:59 PM. You can work in teams of up to five students.

This project contains fewer detailed instructions than the previous assignments. In addition to code, submit a writeup describing your approach, results, and the accuracy of your solution.

While this is still a guided project, it will give you a flavor of research in quantitative economics.

You must decide on grid sizes, numerical methods, and convergence criteria yourself.

Investment subsidy in a heterogeneous firm model

In this project you will study the effects of an investment subsidy in a heterogeneous firm model.

Consider an economy with a continuum of firms indexed by $j \in [0, 1]$. The firms maximize the present discounted value of their profits:

$$\max_{\{i_{j,t}, h_{j,t}\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \left(\frac{1}{1+r} \right)^t \pi_{j,t},$$

where $r > 0$ is the real interest rate. Each firm chooses labor input $h_{j,t}$ and investment $i_{j,t}$. Profits $\pi_{j,t}$ are given by

$$\pi_{j,t} = y_{j,t} - wh_{j,t} - p(i_{j,t})i_{j,t} - c(i_{j,t}, k_{j,t}), \quad y_{j,t} = z_{j,t}k_{j,t}^{\alpha}h_{j,t}^{\nu}.$$

Here $y_{j,t}$ is the output of firm j at time t , which is produced using a production function with productivity $z_{j,t}$, capital $k_{j,t}$, and labor $h_{j,t}$, with $\alpha, \nu \in (0, 1)$ and $\alpha + \nu < 1$. w is the wage rate. The function $p(i_{j,t})$ is the price of investment (which depends on whether the firm is buying or selling capital), and $c(i_{j,t}, k_{j,t})$ is the adjustment cost of investment. Both will be specified later.

The capital stock $k_{j,t}$ evolves according to the standard law of motion:

$$k_{j,t+1} = (1 - \delta)k_{j,t} + i_{j,t},$$

where $\delta \in (0, 1)$ is the depreciation rate of capital.

Finally, the productivity level $z_{j,t}$ follows an AR(1) process in logs

$$\ln z_{j,t+1} = \rho \ln z_{j,t} + (1 - \rho) \ln \tilde{z} + \epsilon_{j,t+1}, \quad \epsilon_{j,t+1} \sim \mathcal{N}(0, \sigma^2),$$

where $\rho \in (0, 1)$ is the persistence parameter, and $\epsilon_{j,t+1}$ is a shock with mean zero and variance σ^2 . $\tilde{z}$ is a constant that normalizes the productivity level: $\mathbb{E} z_{j,t} = 1$.

Empirical moments of firm-level investment

The adjustment cost parameters will be calibrated to match the empirical moments from Cooper and Haltiwanger (2006, *On the Nature of Capital Adjustment Costs*, RES), who study plant-level investment using the Longitudinal Research Database (LRD). The investment rate is $\frac{i_{j,t}}{k_{j,t}}$.

Moment	LRD
Average Investment Rate	12.2%
Inaction Rate ($ i/k < 1\%$)	8.1%
Fraction with Negative Investment	10.4%
Positive Spike Rate ($i/k > 20\%$)	18%
Negative Spike Rate ($i/k < -20\%$)	1.4%

Table 1: Summary Statistics from Cooper and Haltiwanger (2006)

Instructions

To analyze the effects of policy, we first need a model that replicates observed firm behavior. This requires calibrating the model parameters.

1. The optimal labor choice $h_{j,t}^*$ satisfies

$$h_{j,t}^* = \operatorname{argmax}_{h_{j,t} \geq 0} \left(z_{j,t} k_{j,t}^\alpha h_{j,t}^V - w h_{j,t} \right).$$

Find the closed form solution for $h_{j,t}^*$ as a function of $z_{j,t}, k_{j,t}, w$.

Use this to rewrite $\pi_{j,t}$ as a function of $z_{j,t}, k_{j,t}, i_{j,t}$ only.

2. Write down the Bellman equation for the firm's problem, incorporating the functional forms below. Use $V(\cdot)$ to denote the value function.

Adjustment costs:

$$c(i, k) = \frac{\gamma}{2} \left(\frac{i}{k} \right)^2 k + Fk \cdot \mathbb{I}_{\{i \neq 0\}}.$$

The first term is a convex (quadratic) cost penalizing rapid changes; $\gamma > 0$ is its magnitude. The second is a fixed cost proportional to capital, incurred whenever the firm adjusts.

Irreversibilities:

$$p(i) = \mathbb{I}_{\{i \geq 0\}} + p_s \cdot \mathbb{I}_{\{i < 0\}}.$$

Firms pay the full price when purchasing capital but receive only $p_s \leq 1$ when selling.

Eliminating variables that can be solved in closed form is a common technique for reducing complexity in dynamic models.

No need to substitute out investment using the law of motion.

$p_s < 1$ captures partial irreversibility;
 $p_s = 0$ means no resale value.

Calibration

Parameters: $\alpha, \nu, \delta, \rho, \sigma, \tilde{z}, r, w$ (production/productivity) and γ, F, p_s (adjustment costs).

Step 1: External calibration. Set parameters based on standard values:

- $\alpha = 0.30, \nu = 0.60$ (decreasing returns: $\alpha + \nu = 0.90$)
- $\delta = 0.08$ (8% annual depreciation)
- $r = 0.04$ (4% real interest rate)
- $w = 1$ (normalization)
- $\rho = 0.90, \sigma = 0.12$ (productivity process)
- $\tilde{z} = \exp\left(-\frac{\sigma^2}{2(1-\rho^2)}\right)$ to normalize $\mathbb{E}_z = 1$

Step 2: Model exploration. Before estimating parameters, explore the model to verify your numerical solution is accurate.

- Start with $F = 0, p_s = 1$ (only convex costs). The problem is smooth and VFI should converge quickly.
- Introduce $F > 0$ (fixed costs). Regions of inaction should emerge.
- Introduce $p_s < 1$ (irreversibility). Observe asymmetry between positive and negative investment.

Step 3: Method of Moments. Estimate $\theta = (\gamma, F, p_s)$ by matching model moments to the data.

Target moments. Choose which moments from the table to match.

Model moments. For given θ :

1. Solve the Bellman equation to obtain $i^*(k, z; \theta)$.
2. Compute the stationary distribution $\mu(k, z)$.
3. Calculate moments by integrating over μ . For example:

$$\begin{aligned} \text{Inaction rate} &= \int \mathbb{I} \left\{ \left| \frac{i^*}{k} \right| < 0.01 \right\} d\mu, \\ \text{Positive spike rate} &= \int \mathbb{I} \left\{ \frac{i^*}{k} > 0.20 \right\} d\mu. \end{aligned}$$

Estimation. Minimize the distance between model and data moments:

$$\hat{\theta} = \underset{\theta}{\operatorname{argmin}} \left[m^{\text{model}}(\theta) - m^{\text{data}} \right]' W \left[m^{\text{model}}(\theta) - m^{\text{data}} \right],$$

where W is a weighting matrix (e.g., identity).

At each stage: check VFI convergence, test sensitivity to grid size, and verify that policy functions have sensible shapes. Note: the endogenous grid method is not well-suited here due to nonconvexities.

Hint: with fixed costs, compare the value of adjusting ($i \neq 0$) to the value of inaction ($i = 0$) at each state.

Think about which moments are most informative for each parameter. You may use more moments than parameters. Afterwards, you should check whether the model can replicate non-targeted moments as well.

What to report

1. **Model solution:** Bellman equation and numerical method (grids, interpolation, convergence).
2. **Exploration:** Show how policy functions change as you introduce fixed costs and irreversibility (Step 2).
3. **Calibration:** Estimated $(\hat{\gamma}, \hat{F}, \hat{p}_s)$ and moment fit (data vs. model).
4. **Policy functions:** Plot $V(k, z)$ and $i^*(k, z)$ for different z levels.
5. **Stationary distribution:** Distribution over (k, z) and histogram of investment rates i/k .
6. **Accuracy:** VFI convergence and sensitivity to grid size.

Policy analysis: Investment subsidy

Use the calibrated model to analyze the effects of an investment subsidy.

The subsidy. The government subsidizes investment purchases:

$$p(i; \tau) = (1 - \tau) \cdot \mathbb{I}_{\{i \geq 0\}} + p_s \cdot \mathbb{I}_{\{i < 0\}},$$

where $\tau \in [0, 1)$ is the subsidy rate. Firms pay $(1 - \tau)$ when purchasing capital; the resale price p_s is unchanged.

Analysis. Compare the economy with $\tau = 0.10$ (10% subsidy) to baseline ($\tau = 0$). Report:

1. **Policy functions:** How does $i^*(k, z)$ change? Plot for both economies.
2. **Stationary distribution:** How does the distribution over (k, z) change?
3. **Investment moments:** Compute all moments for both economies.
4. **Aggregates:** Compute $K = \int k d\mu$, $H = \int h^* d\mu$, $Y = \int y d\mu$. Report percentage changes.
5. **Cross-sectional patterns:** Compute $\text{Corr}(k, z)$. Are more productive firms larger (higher k)? How does the subsidy affect this correlation?
6. **Subsidy cost:** Compute total cost:

$$\text{Cost} = \tau \int i^*(k, z) \cdot \mathbb{I}_{\{i^* > 0\}} d\mu.$$

Express as a fraction of output. Is the subsidy cost-effective?

7. **Discussion:** Provide economic intuition. How does the subsidy interact with convex costs, fixed costs, and irreversibility?